

# SQL Date & Time Functions

09 · Date & Time Functions — Complete Reference Guide

## Table of Contents

1. GETDATE — Current Date & Time
2. Date Part Extractions (DATETRUNC, DATENAME, DATEPART, YEAR, MONTH, DAY)
3. DATETRUNC — Data Aggregation
4. EOMONTH — End of Month
5. Date Parts Use Cases
6. FORMAT — Date Formatting
7. CONVERT — Type Conversion
8. CAST — Type Casting
9. DATEADD / DATEDIFF — Date Arithmetic
10. ISDATE — Date Validation

## 1. GETDATE() — Current Date & Time

→ Task 1: Display OrderID, CreationTime, a hard-coded date, and today's date

```
SELECT
    OrderID,
    CreationTime,
    '2025-08-20' AS HardCoded,
    GETDATE() AS Today
FROM Sales.Orders;
```

## 2. Date Part Extractions

→ Task 2: Extract various parts of CreationTime

```

SELECT
    OrderID,
    CreationTime,
    DATETRUNC(year,      CreationTime) AS Year_dt,
    DATETRUNC(day,      CreationTime) AS Day_dt,
    DATETRUNC(minute,    CreationTime) AS Minute_dt,
    DATENAME(month,      CreationTime) AS Month_dn,
    DATENAME(weekday,    CreationTime) AS Weekday_dn,
    DATEPART(year,       CreationTime) AS Year_dp,
    DATEPART(month,      CreationTime) AS Month_dp,
    DATEPART(quarter,    CreationTime) AS Quarter_dp,
    YEAR(CreationTime)   AS Year,
    MONTH(CreationTime)  AS Month,
    DAY(CreationTime)    AS Day
FROM Sales.Orders;

```

### 3. DATETRUNC() — Data Aggregation

→ *Task 3: Aggregate orders by year using DATETRUNC*

```

SELECT
    DATETRUNC(year, CreationTime) AS Creation,
    COUNT(*) AS OrderCount
FROM Sales.Orders
GROUP BY DATETRUNC(year, CreationTime);

```

### 4. EOMONTH() — End of Month

→ *Task 4: Display the end-of-month date for each order's CreationTime*

```

SELECT
    OrderID,
    CreationTime,
    EOMONTH(CreationTime) AS EndOfMonth
FROM Sales.Orders;

```

### 5. Date Parts — Use Cases

→ *Task 5: How many orders were placed each year?*

```

SELECT
    YEAR(OrderDate) AS OrderYear,
    COUNT(*) AS TotalOrders
FROM Sales.Orders
GROUP BY YEAR(OrderDate);

```

→ *Task 6: How many orders were placed each month?*

```
SELECT
    MONTH(OrderDate) AS OrderMonth,
    COUNT(*)          AS TotalOrders
FROM Sales.Orders
GROUP BY MONTH(OrderDate);
```

→ *Task 7: Orders per month using friendly month names*

```
SELECT
    DATENAME(month, OrderDate) AS OrderMonth,
    COUNT(*) AS TotalOrders
FROM Sales.Orders
GROUP BY DATENAME(month, OrderDate);
```

→ *Task 8: Show all orders placed in February*

```
SELECT *
FROM Sales.Orders
WHERE MONTH(OrderDate) = 2;
```

## 6. FORMAT() — Date Formatting

→ *Task 9: Format CreationTime into various string representations*

```
SELECT
    OrderID,
    CreationTime,
    FORMAT(CreationTime, 'MM-dd-yyyy') AS USA_Format,
    FORMAT(CreationTime, 'dd-MM-yyyy') AS EURO_Format,
    FORMAT(CreationTime, 'dd')          AS dd,
    FORMAT(CreationTime, 'ddd')          AS ddd,
    FORMAT(CreationTime, 'dddd')         AS dddd,
    FORMAT(CreationTime, 'MM')           AS MM,
    FORMAT(CreationTime, 'MMM')           AS MMM,
    FORMAT(CreationTime, 'MMMM')         AS MMMM
FROM Sales.Orders;
```

→ *Task 10: Custom format — Day Wed Jan Q1 2025 12:34:56 PM*

```
SELECT
    OrderID,
    CreationTime,
    'Day ' + FORMAT(CreationTime, 'ddd MMM') +
    ' Q' + DATENAME(quarter, CreationTime) + ' ' +
    FORMAT(CreationTime, 'yyyy hh:mm:ss tt') AS CustomFormat
FROM Sales.Orders;
```

→ *Task 11: Orders per month formatted as 'Jan 25'*

```
SELECT
    FORMAT(CreationTime, 'MMM yy') AS OrderDate,
    COUNT(*) AS TotalOrders
FROM Sales.Orders
```

```
GROUP BY FORMAT(CreationTime, 'MMM yy');
```

## 7. CONVERT() — Type Conversion

→ *Task 12: Demonstrate CONVERT for type casting and date formatting*

```
SELECT
    CONVERT(INT, '123') AS [String to Int],
    CONVERT(DATE, '2025-08-20') AS [String to Date],
    CreationTime,
    CONVERT(DATE, CreationTime) AS [Datetime to Date],
    CONVERT(VARCHAR, CreationTime, 32) AS [USA Style 32],
    CONVERT(VARCHAR, CreationTime, 34) AS [EURO Style 34]
FROM Sales.Orders;
```

## 8. CAST() — Type Casting

→ *Task 13: Convert data types using CAST*

```
SELECT
    CAST('123' AS INT) AS [String to Int],
    CAST(123 AS VARCHAR) AS [Int to String],
    CAST('2025-08-20' AS DATE) AS [String to Date],
    CAST('2025-08-20' AS DATETIME2) AS [String to Datetime],
    CreationTime,
    CAST(CreationTime AS DATE) AS [Datetime to Date]
FROM Sales.Orders;
```

## 9. DATEADD() / DATEDIFF() — Date Arithmetic

→ *Task 14: Perform date arithmetic on OrderDate*

```
SELECT
    OrderID,
    OrderDate,
    DATEADD(day, -10, OrderDate) AS TenDaysBefore,
    DATEADD(month, 3, OrderDate) AS ThreeMonthsLater,
    DATEADD(year, 2, OrderDate) AS TwoYearsLater
FROM Sales.Orders;
```

→ *Task 15: Calculate the age of employees*

```
SELECT
    EmployeeID,
    BirthDate,
    DATEDIFF(year, BirthDate, GETDATE()) AS Age
FROM Sales.Employees;
```

→ *Task 16: Average shipping duration in days per month*

```

SELECT
    MONTH(OrderDate) AS OrderMonth,
    AVG(DATEDIFF(day, OrderDate, ShipDate)) AS AvgShip
FROM Sales.Orders
GROUP BY MONTH(OrderDate);

```

→ *Task 17: Days between each order and the previous order*

```

SELECT
    OrderID,
    OrderDate AS CurrentOrderDate,
    LAG(OrderDate) OVER (ORDER BY OrderDate) AS PreviousOrderDate,
    DATEDIFF(
        day,
        LAG(OrderDate) OVER (ORDER BY OrderDate),
        OrderDate
    ) AS NrOfDays
FROM Sales.Orders;

```

## 10. ISDATE() — Date Validation

→ *Task 18: Validate dates and replace invalid ones with a default date*

```

SELECT
    OrderDate,
    ISDATE(OrderDate) AS IsValidDate,
    CASE
        WHEN ISDATE(OrderDate) = 1
            THEN CAST(OrderDate AS DATE)
        ELSE '9999-01-01'
    END AS NewOrderDate
FROM (
    SELECT '2025-08-20' AS OrderDate UNION
    SELECT '2025-08-21' UNION
    SELECT '2025-08-23' UNION
    SELECT '2025-08' -- Invalid
) AS t

```